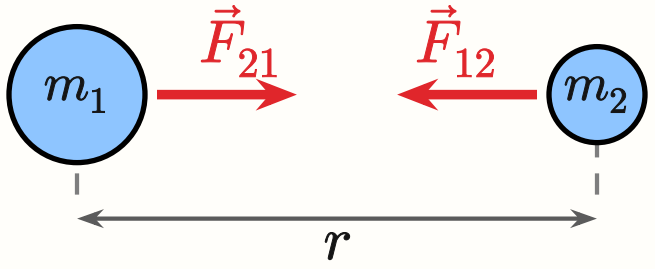
Newton's Law of Gravitation

Class 11 · examples use $G = 6.67 \times 10^{-11}\,\mathrm{N\,m^2 kg^{-2}}$, $g = 9.8\,\mathrm{m\,s^{-2}}$ (surface), Earth $R = 6.4 \times 10^{6}\,\mathrm{m}$, $M = 6 \times 10^{24}\,\mathrm{kg}$

1 The law



$$F = \frac{G m_1 m_2}{r^2}$$

POINT masses, or spherically symmetric bodies with r between CENTRES · always attractive, along the line joining them

$$\vec{F}_{12} = -\frac{G m_1 m_2}{r^2}\,\hat{r}_{12}$$

$\vec{F}_{12}$: force on m_2 due to m_1 · $\hat{r}_{12}$: unit vector from m_1 to m_2 · $\vec{F}_{21} = -\vec{F}_{12}$ (Newton's third law)

2 Universal constant G

- $G = 6.67 \times 10^{-11}\,\mathrm{N\,m^2 kg^{-2}}$, *dimensions* $[\mathrm{M^{-1}L^3T^{-2}}]$; *same everywhere, does NOT depend on the medium between the masses.*
- *Central and conservative force; weakest of the fundamental forces, but long range and never screened.*

Example

Two $100\,\mathrm{kg}$ *spheres, centres* $1\,\mathrm{m}$ *apart:*

$$F = 6.67 \times 10^{-11} \times \frac{100 \times 100}{1^2} = 6.67 \times 10^{-7}\,\mathrm{N}$$ *(tiny).*

Syllabus Checklist

official JEE Main + JEE Advanced syllabus + standard results

1 JEE Main + JEE Advanced syllabus

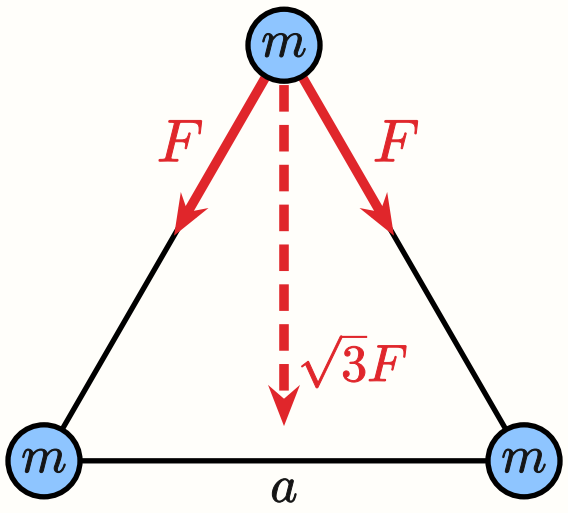
- ✓ Universal law of gravitation, G, vector form — p1, p3
- ✓ Kepler's laws of planetary motion — p4, p5
- ✓ Acceleration due to gravity g — p6
- ✓ Variation of g with altitude — p7
- ✓ Variation of g with depth — p8
- ✓ Variation of g with latitude and rotation — p9
- ✓ Gravitational field (intensity) — p10, p11
- ✓ Gravitational potential — p12, p13
- ✓ Gravitational potential energy — p14, p15
- ✓ Escape speed — p16, p17
- ✓ Motion of satellites: orbital speed, time period — p18
- ✓ Energy of a satellite — p19
- ✓ Geostationary and polar satellites — p20
- ✓ Weightlessness — p21

2 Standard JEE results also covered

- ✓ Superposition, Cavendish, inertial vs gravitational mass — p3
- ✓ Areal velocity and angular momentum conservation — p4, p5
- ✓ Field and potential: ring, rod, shell, solid sphere — p10–p13
- ✓ Self energy and binding energy — p15, p19
- ✓ Orbit shape vs launch speed; transfer between orbits — p22, p23
- ✓ Cavity problems (superposition) — p24
- ✓ Binary stars, two-body orbit, reduced mass — p25
- ✓ Tunnel through the Earth (SHM) — p26
- ✓ Traps, results table, solved questions — p27–p33

Superposition and Measuring G

1 Superposition of forces



$$\vec{F} = \vec{F}_1 + \vec{F}_2 + \dots$$

VECTOR sum; each pair acts as if the others were absent

Example

Three equal masses m at the corners of an equilateral triangle of side a: each feels $F = Gm^2/a^2$ from the other two at $60°$; net $\sqrt{3}\,Gm^2/a^2$ towards the centroid.

2 Cavendish experiment

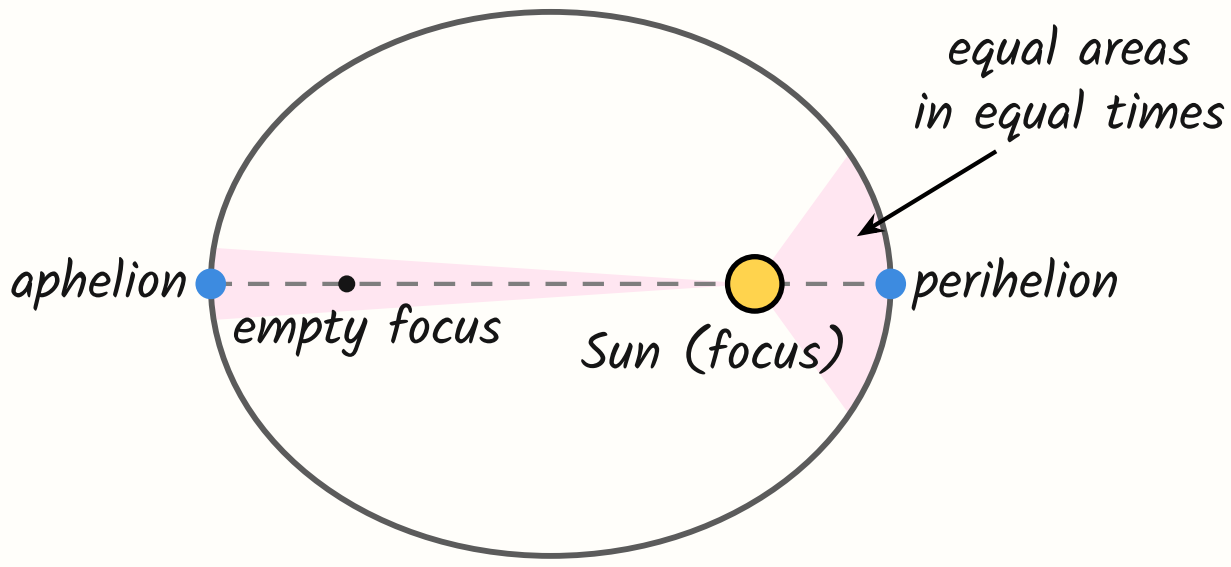
- *Torsion balance: small spheres on a light rod, large spheres brought near; the attraction twists the wire. At equilibrium (torsion constant k, rod length L, centre gap d): $k\theta = \dfrac{GMm}{d^2}L$ gives G.*
- *After G is known: Earth's mass $M = gR^2/G$ (weighing the Earth, p6).*

3 Two kinds of mass

- *Inertial mass: $m = F/a$. Gravitational mass: $m = F_{grav}/g$. Experiments show they are EQUAL, so all bodies fall with the same g (no air drag).*

Kepler's Laws

1 Law of orbits



- Every planet moves in an ELLIPSE with the Sun at one FOCUS (circle: special case $e = 0$).

2 Law of areas

$$\frac{dA}{dt} = \frac{L}{2m} = \text{constant}$$

gravity is a CENTRAL force: torque about the Sun $= 0$, so $L = mrv_{\perp}$ is conserved · faster near the Sun

3 Law of periods

$$T^2 = \frac{4\pi^2}{GM} a^3 \quad (T^2 \propto a^3)$$

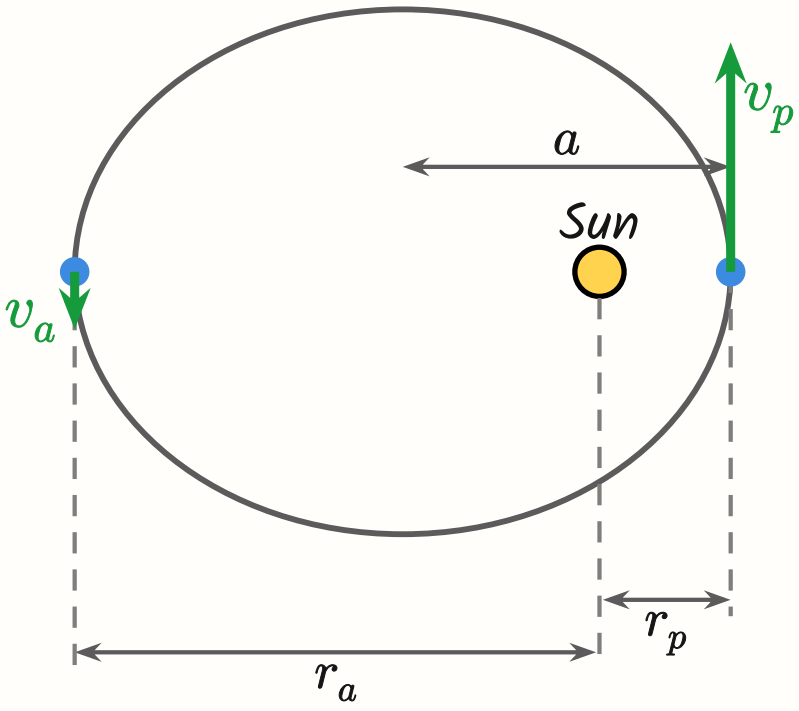
a: semi-major axis · M: Sun's mass, planet mass $m \ll M$ (else $G(M+m)$) · circle: $a = r$

Example

A planet with $a = 4\,\text{AU}$ (Earth: $1\,\text{AU}$, $1\,\text{yr}$): $T = \sqrt{4^3} = 8\,\text{yr}$.

Elliptical Orbits

1 Geometry and speeds



$$r_p = a(1-e), \quad r_a = a(1+e)$$

$$\frac{v_p}{v_a} = \frac{r_a}{r_p} = \frac{1+e}{1-e}$$

e: eccentricity · L conserved: $mv_p r_p = mv_a r_a$ (velocity $\perp$ radius ONLY at these two points)

2 Energy in an elliptical orbit

$$E = -\frac{GMm}{2a}, \qquad v^2 = GM\left(\frac{2}{r} - \frac{1}{a}\right)$$

$m \ll M$, M fixed at the focus · E depends only on a · second form (vis-viva): speed at ANY distance r

Example

$e = 0.6$: $v_p/v_a = 1.6/0.4 = 4$. Earth ($e \approx 0.0167$): $v_p/v_a \approx 1.034$, nearly a circle.

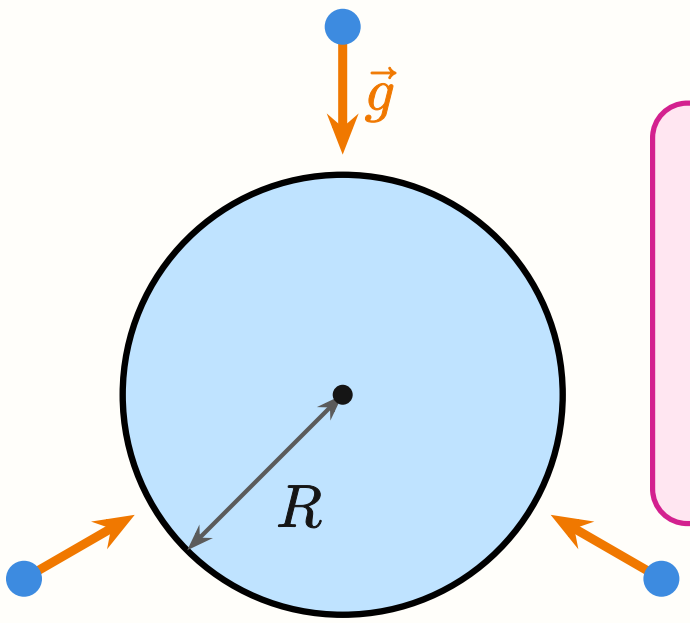
Golden points

★ L, E and dA/dt are constant along the whole orbit; K and U keep changing (K largest at perihelion).

★ $v_p = \sqrt{\dfrac{GM}{a} \cdot \dfrac{1+e}{1-e}}$ (vis-viva at $r = r_p$).

Acceleration due to Gravity

1 g at the surface



$$g = \frac{GM}{R^2} = \frac{4}{3}\pi G\rho R$$

uniform (or spherically symmetric) Earth, NON-rotating · ρ: mean density · $\vec{g}$ towards the centre

Example

Mass of the Earth: $M = \dfrac{gR^2}{G} = \dfrac{9.8 \times (6.4 \times 10^6)^2}{6.67 \times 10^{-11}} \approx 6.0 \times 10^{24}\,\mathrm{kg}$;

mean density $\rho = \dfrac{3g}{4\pi GR} \approx 5.5 \times 10^3\,\mathrm{kg\,m^{-3}}$.

---

2 Another planet

$$\frac{g_p}{g} = \frac{M_p}{M}\left(\frac{R}{R_p}\right)^2 = \frac{\rho_p}{\rho}\cdot\frac{R_p}{R}$$

both spherical, uniform, non-rotating

Example

Planet with twice the Earth's radius and the same density: $g_p = 2g = 19.6\,\mathrm{m\,s^{-2}}$. *Same mass, half the radius:* $4g$.

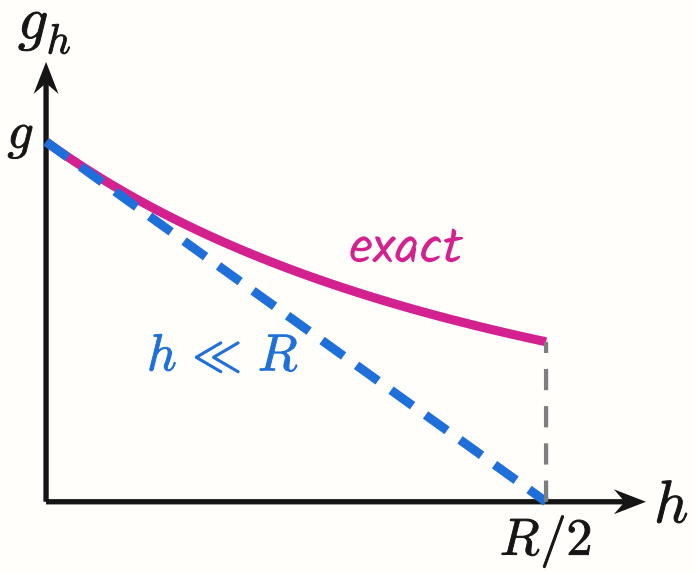
g at a Height

1 Exact result

$$g_h = \frac{GM}{(R+h)^2} = \frac{g}{\left(1+\dfrac{h}{R}\right)^2}$$

uniform spherical Earth, ignoring rotation · h above the SURFACE · valid for ANY h

2 Small heights



$$g_h \approx g\left(1-\frac{2h}{R}\right)$$

ONLY for $h \ll R$ (binomial) · fractional fall $\dfrac{\Delta g}{g} = \dfrac{2h}{R}$

Example

$h = 32\,\text{km}$ $(h/R = 0.005)$: *exact* $g_h = \dfrac{9.8}{1.005^2} \approx 9.703\,\text{m s}^{-2}$;
approximate $9.8 \times 0.99 = 9.702\,\text{m s}^{-2}$ *(a 1% fall).*

Golden points

★ $g_h = g/4$ *at* $h = R$ *(exact formula:* $(1+1)^2 = 4$*). The approximation would give* $-g$*: never use it for large* h*.*

★ $g_h \propto 1/r^2$ *with* $r = R + h$ *from the CENTRE.*

g at a Depth

1 Inside the Earth

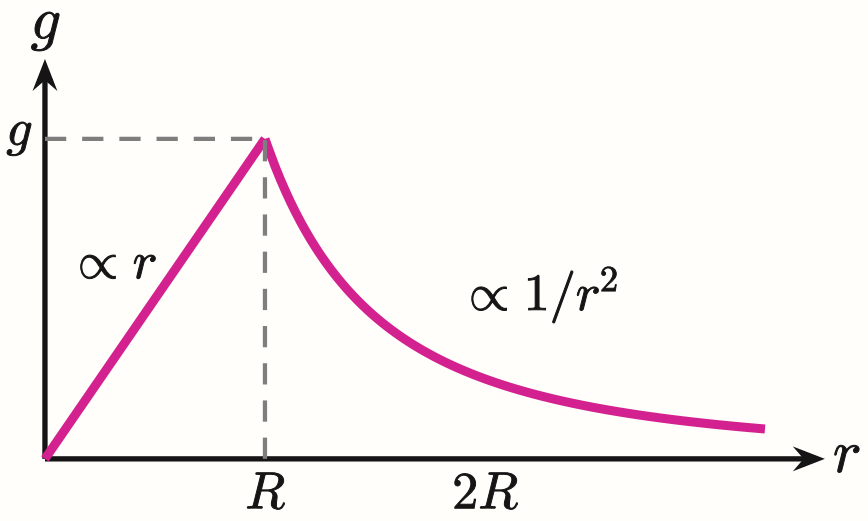
$$g_d = g\left(1 - \frac{d}{R}\right) = \frac{GM}{R^3}r$$

UNIFORM-density Earth, ignoring rotation · $r = R - d$ from the centre · the outer shell above the point pulls with ZERO net force (shell theorem)

Example

$d = 32\,\text{km}$: $g_d = 9.8 \times (1 - 0.005) \approx 9.751\,\text{m s}^{-2}$ *(only* 0.5% *less).*
At $d = R/2$: $g/2 = 4.9\,\text{m s}^{-2}$; *centre:* 0.

2 g against r



- *Linear inside (uniform Earth), maximum AT the surface,* $1/r^2$ *outside.*
- *At* $2R$ *from the centre:* $g/4$; *at* $R/2$: $g/2$.

3 Height or depth?

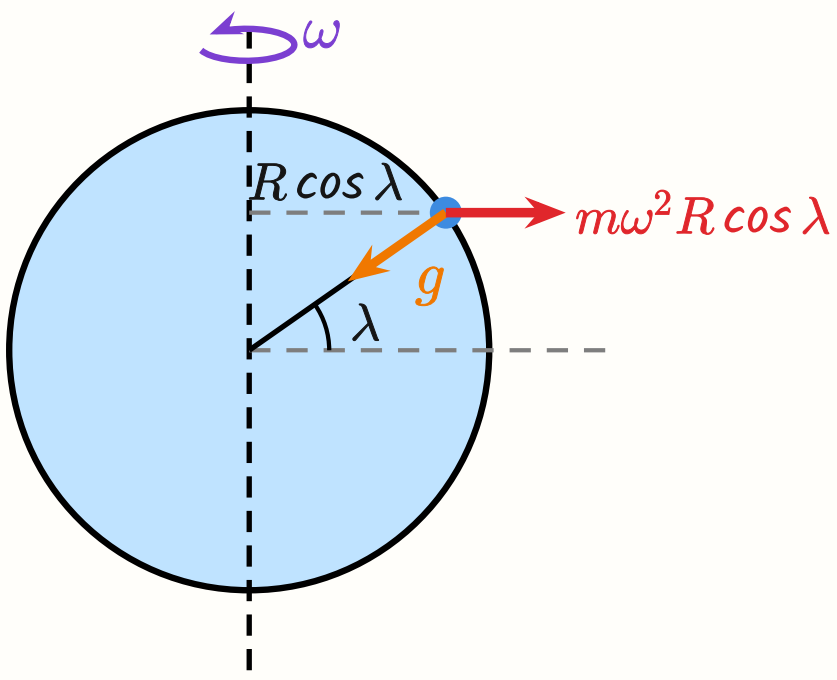
- *Same small* $h = d$ $(\ll R)$: $\Delta g_{height} = 2gh/R$ *is TWICE* $\Delta g_{depth} = gd/R$. *Equal* g *for* $d = 2h$.

Golden points

★ g *decreases BOTH ways from the surface (uniform Earth). The real Earth has a denser core, so* g_d *is not linear.*

g with Latitude and Rotation

1 Effect of the Earth's spin



$$g' = g - \omega^2 R \cos^2 \lambda$$

λ: latitude · spherical uniform Earth, $\omega^2 R \ll g$ · g': effective (apparent) g · red: centrifugal force (Earth's frame)

- Poles ($\lambda = 90°$): $g' = g$ (maximum). Equator ($\lambda = 0$): $g' = g - \omega^2 R$ (minimum).
- $\omega = \dfrac{2\pi}{86400\,\mathrm{s}} \approx 7.27 \times 10^{-5}\,\mathrm{rad\,s^{-1}}$; $\omega^2 R \approx 0.034\,\mathrm{m\,s^{-2}}$, so at the equator $g' \approx 9.766\,\mathrm{m\,s^{-2}}$; at $\lambda = 60°$ the fall is $\approx 0.0085\,\mathrm{m\,s^{-2}}$.

2 Weightless at the equator

$$\omega = \sqrt{\frac{g}{R}} \approx 1.24 \times 10^{-3}\,\mathrm{rad\,s^{-1}}$$

$g' = 0$ at the equator · day would last $2\pi\sqrt{R/g} \approx 84.6\,\mathrm{min}$, about 17 times faster spin

Golden points

★ Real Earth is flattened at the poles ($R_{pole} < R_{equator}$): this ALSO makes g larger at the poles.

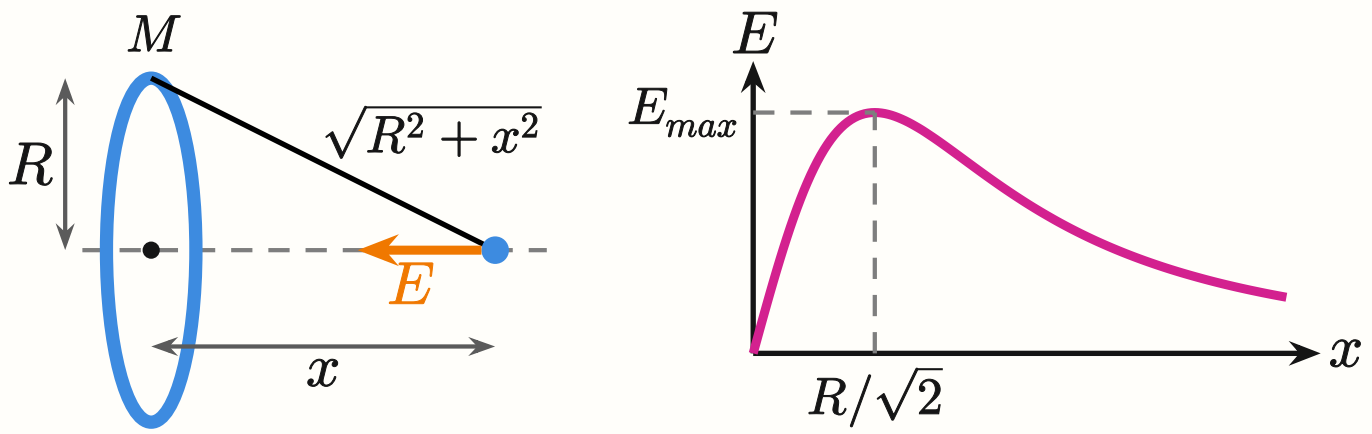
Gravitational Field

1 Definition; point mass

$$\vec{E} = \frac{\vec{F}}{m} = -\frac{GM}{r^2}\hat{r}$$

force per unit mass on a small test mass · point mass (or outside a spherically symmetric body) · unit $\mathrm{N\,kg^{-1}} = \mathrm{m\,s^{-2}}$ *· towards the mass*

2 Ring on its axis



$$E = \frac{GMx}{(R^2 + x^2)^{3/2}}, \qquad E_{max} = \frac{2GM}{3\sqrt{3}R^2} \text{ at } x = \frac{R}{\sqrt{2}}$$

thin UNIFORM ring, point on its AXIS · towards the centre · $E = 0$ *at the centre ·* $x \gg R$*:* GM/x^2

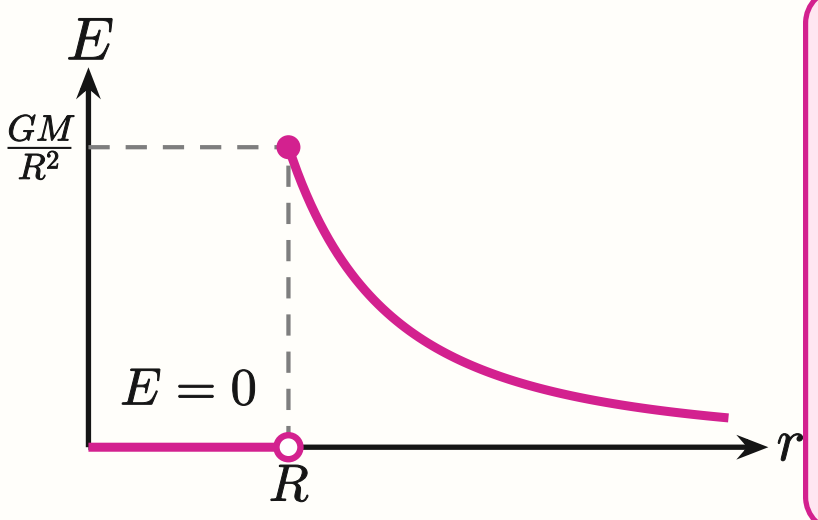
3 Uniform rod on its axis

$$E = \frac{GM}{d(d + L)}$$

thin uniform rod, mass M*, length* L *· point on its line at distance* d *from the near end*

Field of a Shell and a Sphere

1 Uniform thin shell

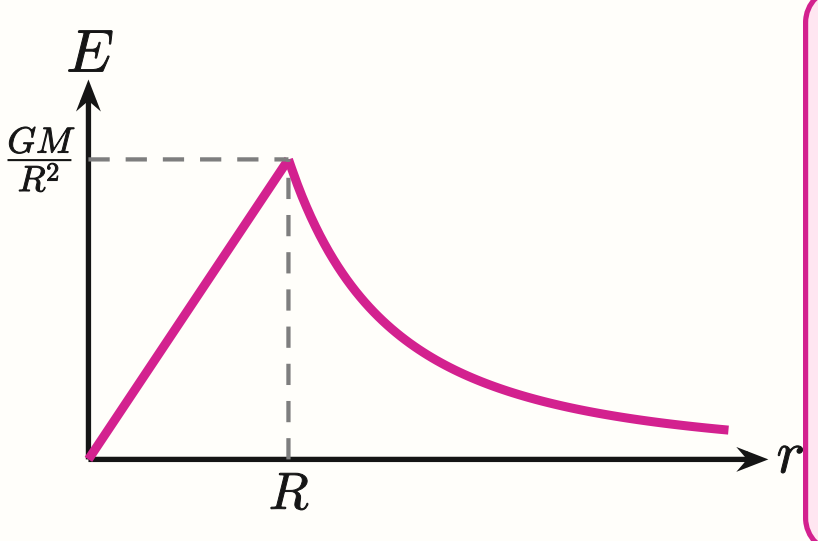


$$E = 0 \quad (r < R)$$

$$E = \frac{GM}{r^2} \quad (r > R)$$

shell theorem: UNIFORM thin shell · outside: as if all mass at the centre · jumps at $r = R$

2 Uniform solid sphere



$$E = \frac{GMr}{R^3} \quad (r \le R)$$

$$E = \frac{GM}{r^2} \quad (r \ge R)$$

UNIFORM density · inside: only the mass within r pulls · maximum AT the surface

Example

Uniform solid sphere: $E(R/2) = GM/2R^2$, $E(2R) = GM/4R^2$, so $E(R/2) = 2E(2R)$.

Golden points

- ★ Inside a uniform shell: NO force from the shell ANYWHERE, not only at the centre.
- ★ Two equal fields: one inside (r_1) and one outside (r_2) a uniform solid sphere when $r_1 r_2^2 = R^3$.

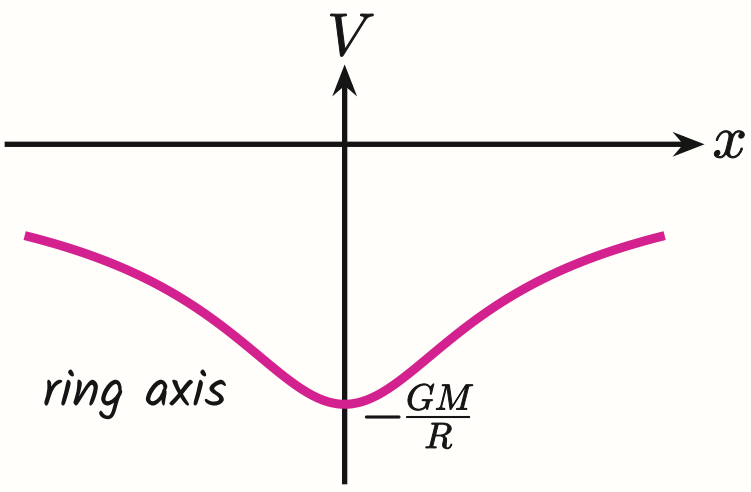
Gravitational Potential

1 Definition; point mass

$$V = \frac{W_{\infty \to P}}{m} = -\frac{GM}{r}$$

work by an EXTERNAL agent per unit mass, bringing a test mass SLOWLY from infinity · $V = 0$ at infinity · SCALAR, always negative · $\mathrm{J\,kg^{-1}}$

2 Ring and rod



$$V_{ring} = -\frac{GM}{\sqrt{R^2 + x^2}}$$

point on the axis · centre: $-GM/R$ (even if the mass is NOT uniform)

$$V_{rod} = -\frac{GM}{L} \ln \frac{d + L}{d}$$

uniform rod, point on its line, d from the near end

3 Field from potential

$$E_r = -\frac{dV}{dr}, \qquad \vec{E} = -\left(\frac{\partial V}{\partial x}\hat{i} + \frac{\partial V}{\partial y}\hat{j} + \frac{\partial V}{\partial z}\hat{k}\right)$$

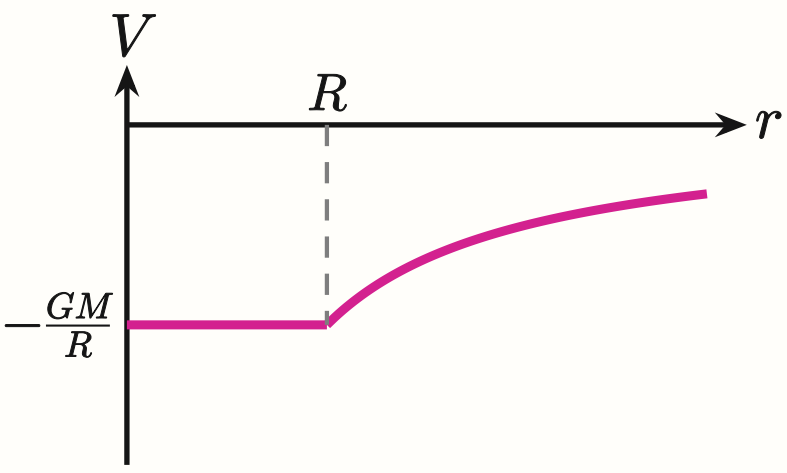
$\vec{E}$ points towards DECREASING V · $V_B - V_A = -\int_A^B \vec{E} \cdot d\vec{r}$

Example

$V = (2x^2 + 3y)\,\mathrm{J\,kg^{-1}}$ *(x, y in m)*: $\vec{E} = -(4x\,\hat{i} + 3\,\hat{j})$; *at* $(1, 1)\,\mathrm{m}$: $|\vec{E}| = 5\,\mathrm{N\,kg^{-1}}$; *force on* $2\,\mathrm{kg}$: $10\,\mathrm{N}$.

Potential of a Shell and a Sphere

1 Uniform thin shell

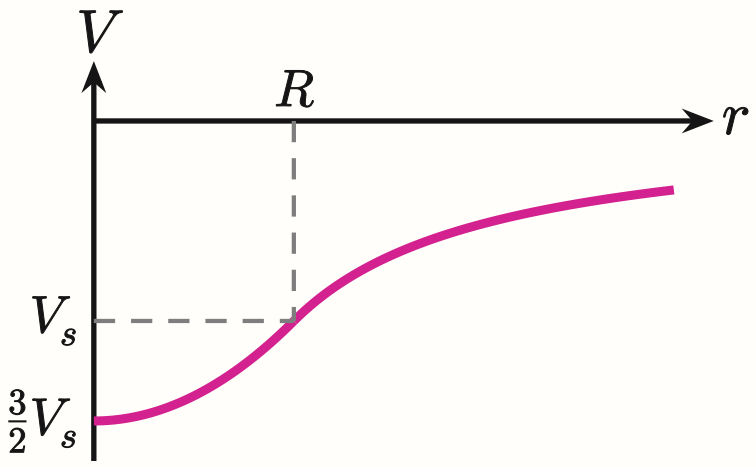


$$V = -\frac{GM}{R} \quad (r \le R)$$

$$V = -\frac{GM}{r} \quad (r \ge R)$$

UNIFORM thin shell · inside: V constant (so $E = 0$), NOT zero

2 Uniform solid sphere



$$V = -\frac{GM(3R^2 - r^2)}{2R^3}$$

inside, $r \le R$ · UNIFORM density · outside $-GM/r$ · centre $-\frac{3GM}{2R} = 1.5\,V_s$ (lowest point of the well)

Example

Earth as a uniform sphere: $V_s = -gR = -6.27 \times 10^7\ \mathrm{J\,kg^{-1}}$; centre $-9.41 \times 10^7\ \mathrm{J\,kg^{-1}}$. Work to take $1\,\mathrm{kg}$ SLOWLY from the centre to the surface: $\frac{1}{2}gR \approx 3.14 \times 10^7\ \mathrm{J}$.

Golden points

★ *$E = 0$ does NOT mean $V = 0$: inside a shell $E = 0$ but $V = -GM/R$. V is continuous everywhere; E jumps only at a thin shell.*

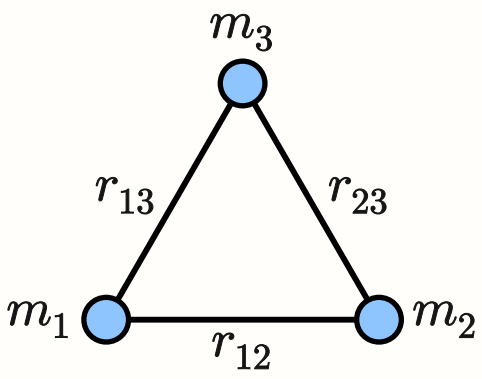
Gravitational Potential Energy

1 Two masses

$$U = -\frac{Gm_1m_2}{r}$$

point (or spherically symmetric) masses · $U = 0$ at infinite separation · always NEGATIVE (bound) · $U = mV$

2 System of masses



$$U = -G\left(\frac{m_1m_2}{r_{12}} + \frac{m_2m_3}{r_{23}} + \frac{m_1m_3}{r_{13}}\right)$$

one term per PAIR (n masses: $n(n-1)/2$ pairs)

Example

Three $1\,\mathrm{kg}$ masses on an equilateral triangle of side $1\,\mathrm{m}$: $U = -3G \approx -2.0 \times 10^{-10}\,\mathrm{J}$; minimum work to pull them apart to infinity: $+2.0 \times 10^{-10}\,\mathrm{J}$.

3 Lifting a body from the surface

$$\Delta U = \frac{GMmh}{R(R+h)} = \frac{mgh}{1+h/R} \approx mgh$$

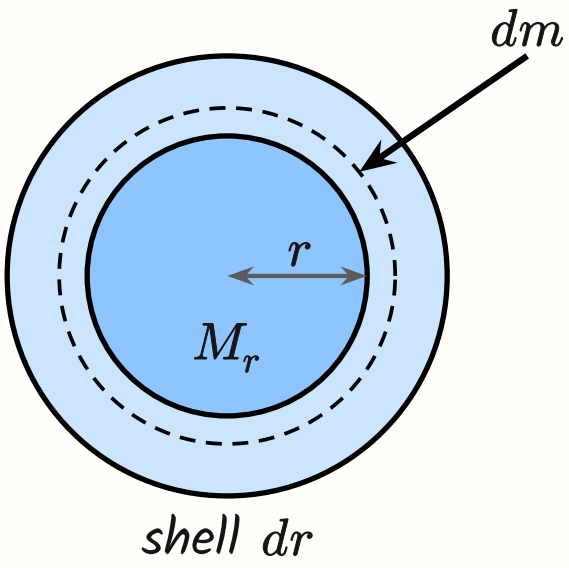
uniform spherical Earth · mgh ONLY for $h \ll R$ · lifting to $h = R$: $\frac{1}{2}mgR$

Example

$100\,\mathrm{kg}$ to $h = R$: $\Delta U = \frac{1}{2}mgR \approx 3.14 \times 10^9\,\mathrm{J}$ (mgh gives double).

Self Energy and Binding Energy

1 Building a body from dust



$$U_{shell} = -\frac{GM^2}{2R}$$

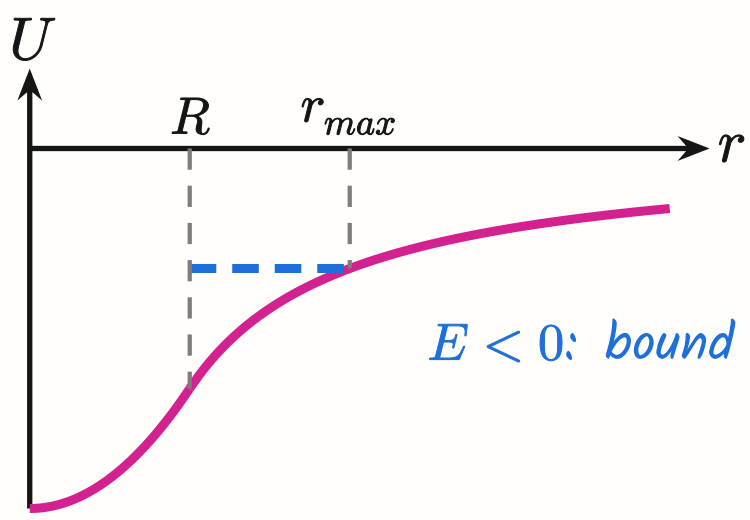
$$U_{sphere} = -\frac{3GM^2}{5R}$$

SELF energy: UNIFORM thin shell; UNIFORM solid sphere · each dm brought from infinity · $U = 0$ when spread to infinity

2 Binding energy

- *Binding energy = minimum energy to pull a bound system apart to infinity = $-E_{total}$ (for $E < 0$).*
- *Uniform sphere: $3GM^2/5R$. Circular-orbit satellite: $GMm/2r$ (p19). Body at rest on the surface (ignoring rotation): GMm/R.*

3 Potential well



- *Uniform Earth: $-1/r$ outside, parabola inside (deepest at the centre).*
- *Total energy $E < 0$: turns back at r_{max} where $U = E$ (no drag).*

Escape Speed

1 From the surface

$$\tfrac{1}{2}mv_e^2 - \frac{GMm}{R} = 0 \;\Rightarrow\; v_e = \sqrt{\frac{2GM}{R}} = \sqrt{2gR}$$

least launch speed to reach infinity (speed $\to 0$ there) · no air drag, ignoring Earth's rotation and other bodies

Example

Earth: $v_e = \sqrt{2 \times 9.8 \times 6.4 \times 10^6} = 11.2\,\mathrm{km\,s^{-1}}$. *Moon* ($M = 7.35 \times 10^{22}\,\mathrm{kg}$, $R = 1.74 \times 10^6\,\mathrm{m}$): $v_e \approx 2.37\,\mathrm{km\,s^{-1}}$.

2 Key properties

- *Independent of the body's mass and of the launch DIRECTION (if the path does not hit the planet).*
- *From height h:* $v_e = \sqrt{\dfrac{2GM}{R+h}}$; *at $h = R$:* $11.2/\sqrt{2} \approx 7.92\,\mathrm{km\,s^{-1}}$.
- *Same density ρ:* $v_e = R\sqrt{\frac{8}{3}\pi G\rho} \propto R$. *Same mass:* $v_e \propto 1/\sqrt{R}$.
- *Small v_e (Moon): fast gas molecules leak away over time, so no atmosphere.*

Golden points

★ *Rotating Earth: launching EASTWARD at the equator saves $\omega R \approx 0.465\,\mathrm{km\,s^{-1}}$.*

Projection from the Earth

1 Launched up with $v < v_e$

$$h_{max} = \frac{Rv^2}{2gR - v^2}$$

spherical Earth, vertical launch, no air drag, ignoring rotation · energy: $\frac{1}{2}mv^2 - \frac{GMm}{R} = -\frac{GMm}{R+h}$

Example

$v = v_e/2$ $(v^2 = gR/2)$: $h_{max} = \dfrac{R \cdot gR/2}{2gR - gR/2} = \dfrac{R}{3} \approx 2133\,\mathrm{km}$. *The flat-Earth formula* $v^2/2g$ *gives only* $R/4 = 1600\,\mathrm{km}$: *WRONG for large* v.

- $v = \sqrt{gR} \approx 7.92\,\mathrm{km\,s^{-1}}$ *straight up:* $h_{max} = R$.
- *For* $v \ll v_e$: $h_{max} \approx v^2/2g$ *(constant* g*).*

2 Launched with $v > v_e$

$$v_\infty = \sqrt{v^2 - v_e^2}$$

speed far away (energy conservation, no drag) · any direction that does not hit the planet

Example

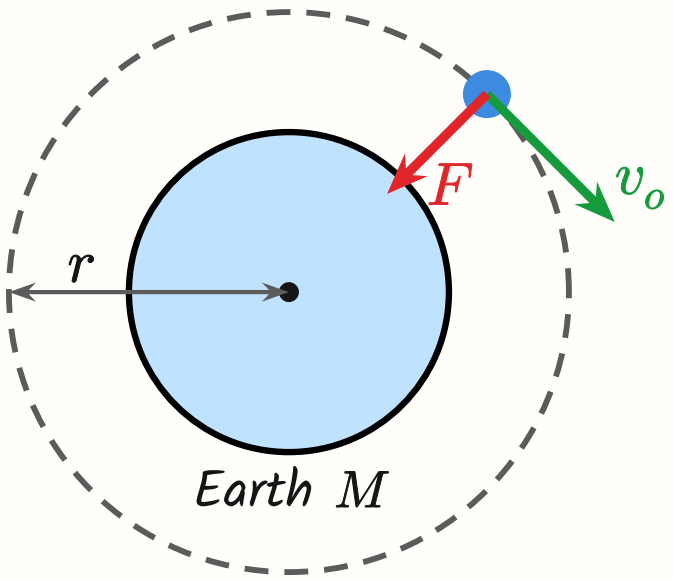
$v = 2v_e = 22.4\,\mathrm{km\,s^{-1}}$: $v_\infty = \sqrt{3}\,v_e \approx 19.4\,\mathrm{km\,s^{-1}}$.

Golden points

★ *Use* $U = -GMm/r$ *(never* mgh*) whenever* h *is comparable to* R.

Orbital Speed and Period

1 Circular orbit



$$\frac{mv_o^2}{r} = \frac{GMm}{r^2} \quad \Rightarrow \quad v_o = \sqrt{\frac{GM}{r}}$$

CIRCULAR orbit, $r = R + h$ from the centre, satellite $m \ll M$, gravity the only force · independent of m

$$T = \frac{2\pi r}{v_o} = 2\pi\sqrt{\frac{r^3}{GM}}, \qquad L = m\sqrt{GMr}$$

$T^2 \propto r^3$ (Kepler's third law) · $GM = gR^2$ for the Earth

2 Standard values (Earth)

- *Just above the surface ($h \ll R$):* $v_o = \sqrt{gR} \approx 7.92\,\mathrm{km\,s^{-1}}$, $T = 2\pi\sqrt{R/g} \approx 84.6\,\mathrm{min}$.
- $h = R$: $v_o = 7.92/\sqrt{2} = 5.6\,\mathrm{km\,s^{-1}}$, $T = 2\sqrt{2} \times 84.6 \approx 239\,\mathrm{min}$.

Example

Space station at $h = 400\,\mathrm{km}$ ($r = 6.8 \times 10^6\,\mathrm{m}$):

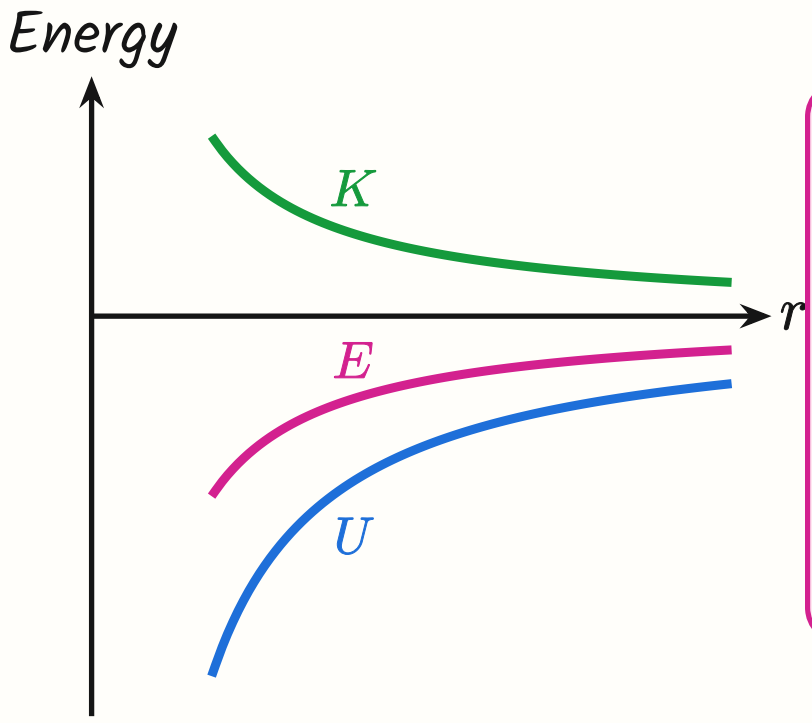
$$v_o = \sqrt{\frac{gR^2}{r}} \approx 7.68\,\mathrm{km\,s^{-1}}, \; T \approx 92.7\,\mathrm{min}.$$

Golden points

★ *Higher orbit: SLOWER ($v \propto r^{-1/2}$) and LONGER period ($T \propto r^{3/2}$).*

Energy of a Satellite

1 K, U and E



$$K = \frac{GMm}{2r}, \quad U = -\frac{GMm}{r}$$

$$E = K + U = -\frac{GMm}{2r}$$

CIRCULAR orbit, $U = 0$ at infinity · $K = -E$, $U = 2E$

Example

$m = 1000\,\text{kg}$ at $h = R$ $(r = 2R)$: $E = -\dfrac{mgR}{4} \approx -1.57 \times 10^{10}\,\text{J}$, $K = +1.57 \times 10^{10}\,\text{J}$, $U = -3.14 \times 10^{10}\,\text{J}$.

2 Binding energy and transfers

$$BE = \frac{GMm}{2r}, \qquad \Delta E_{r_1 \to r_2} = \frac{GMm}{2}\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$$

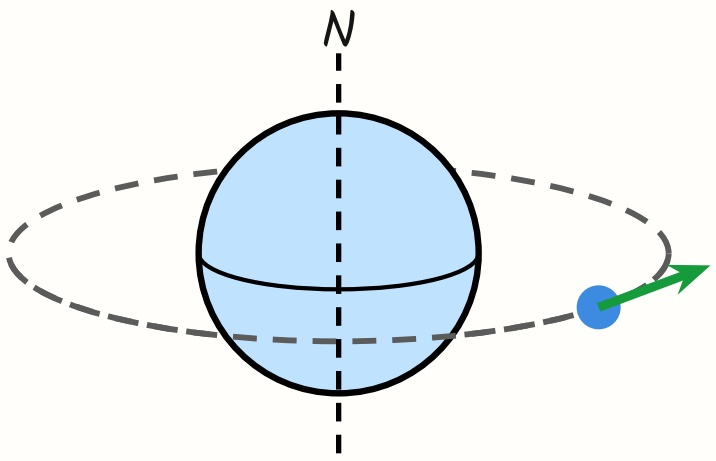
circular orbits · launch from rest on the surface (non-rotating Earth, no drag) to orbit r: $GMm\left(\frac{1}{R} - \frac{1}{2r}\right)$

Golden points

★ Air drag (slow loss of E, nearly circular orbit): E falls, r shrinks, so $K = -E$ RISES: the satellite speeds up.

Geostationary and Polar Satellites

1 Geostationary satellite



geostationary orbit in the equatorial plane (not to scale)

- Circular orbit in the EQUATORIAL plane, moving WEST to EAST.
- T = Earth's rotation period (24 h; strictly one sidereal day $\approx 23\,\text{h}\ 56\,\text{min}$): stays above ONE point.

$$r = \left(\frac{gR^2T^2}{4\pi^2}\right)^{1/3} \approx 4.23 \times 10^7\,\text{m} \approx 6.6R$$

$T = 86400\,\text{s}$, $GM = gR^2$ · height $h = r - R \approx 3.59 \times 10^7\,\text{m} \approx 36000\,\text{km}$

- Uses: communication, TV broadcast, weather. Same r for EVERY geostationary satellite, any mass.

2 Polar satellite

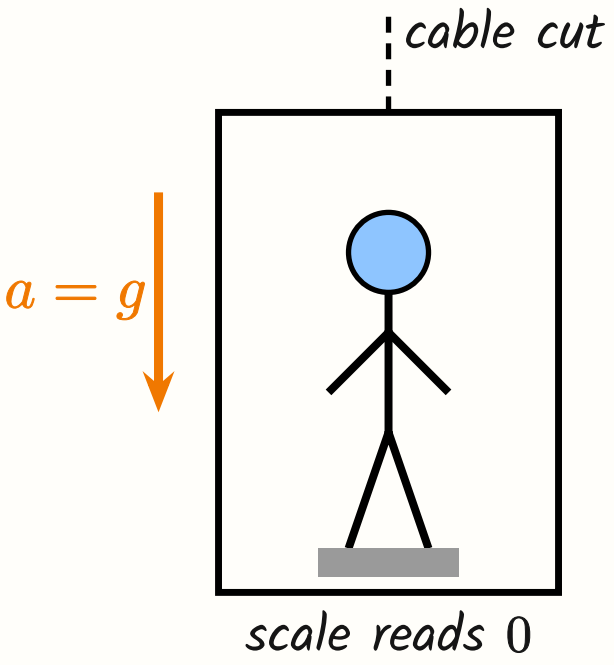
- Low orbit ($h \approx 500$–$800\,\text{km}$) passing over both POLES; $T \approx 95$–$101\,\text{min}$ (circular orbit, $T = 2\pi\sqrt{r^3/gR^2}$).
- Earth spins beneath it, so it scans the whole surface strip by strip: remote sensing, mapping, weather. Sun-synchronous ones cross a place at the same local time.

Golden points

★ A satellite whose period is $24\,\text{h}$ but whose orbit is NOT equatorial is geosynchronous, not geostationary.

Weightlessness

1 Apparent weight



$$N = m(g - a)$$

lift (or frame) accelerating DOWN with $a \leq g$ · N: scale reading (apparent weight) · up: $m(g + a)$

$$a = g \;\Rightarrow\; N = 0$$

free fall: WEIGHTLESS

2 Weightless in orbit

- Astronaut and spacecraft both fall freely with the SAME acceleration g_h: no contact force, apparent weight zero.
- Gravity is NOT zero there: at $h = 400\,\mathrm{km}$ (spherical Earth), $g_h = 9.8 \times (6.4/6.8)^2 \approx 8.68\,\mathrm{m\,s^{-2}}$ ($\approx 89\%$ of g).
- Mass and inertia are unchanged.

3 Inside an orbiting satellite

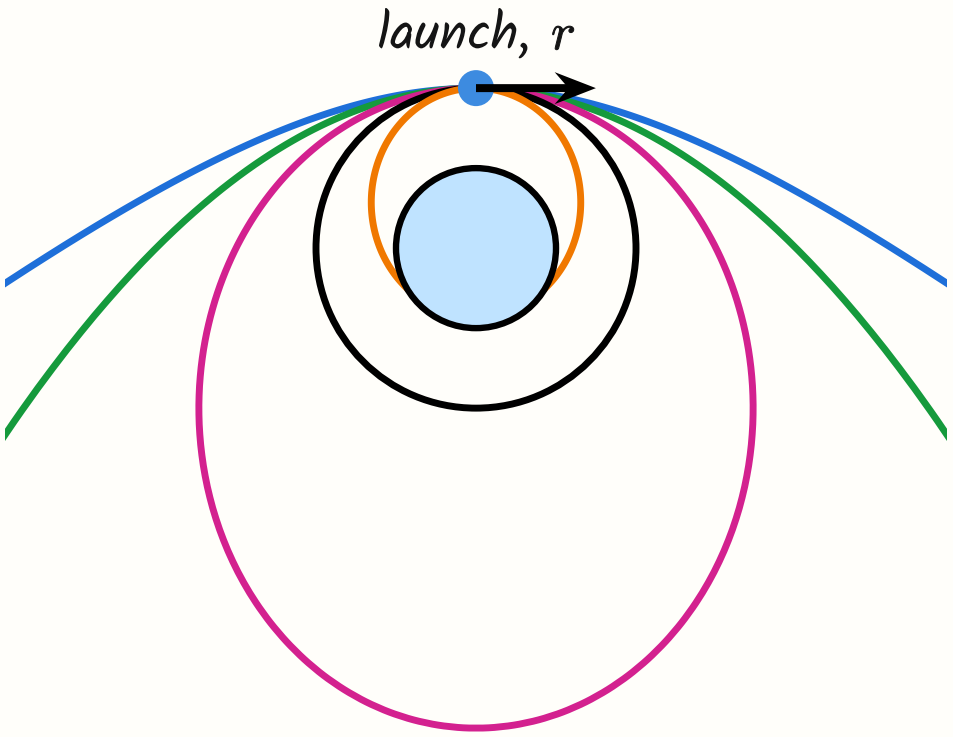
- Spring balance reads 0; a simple pendulum does not oscillate ($g_{eff} = 0$, $T \to \infty$); water does not pour; a candle flame is spherical (no convection).

Golden points

★ Weightless means APPARENT weight zero (free fall), not mass zero and not gravity zero.

Launch Speed and Orbit Shape

1 Tangential launch at distance r



- $v < v_o$: ellipse, launch point = APOGEE (may hit the Earth)
- $v = v_o = \sqrt{GM/r}$: circle
- $v_o < v < \sqrt{2}v_o$: ellipse, launch point = PERIGEE
- $v = \sqrt{2}v_o$: parabola (just escapes)
- $v > \sqrt{2}v_o$: hyperbola

2 Energy decides

$$E = \tfrac{1}{2}mv^2 - \frac{GMm}{r}$$

$E < 0$: **bound**, $E \geq 0$: **escapes**

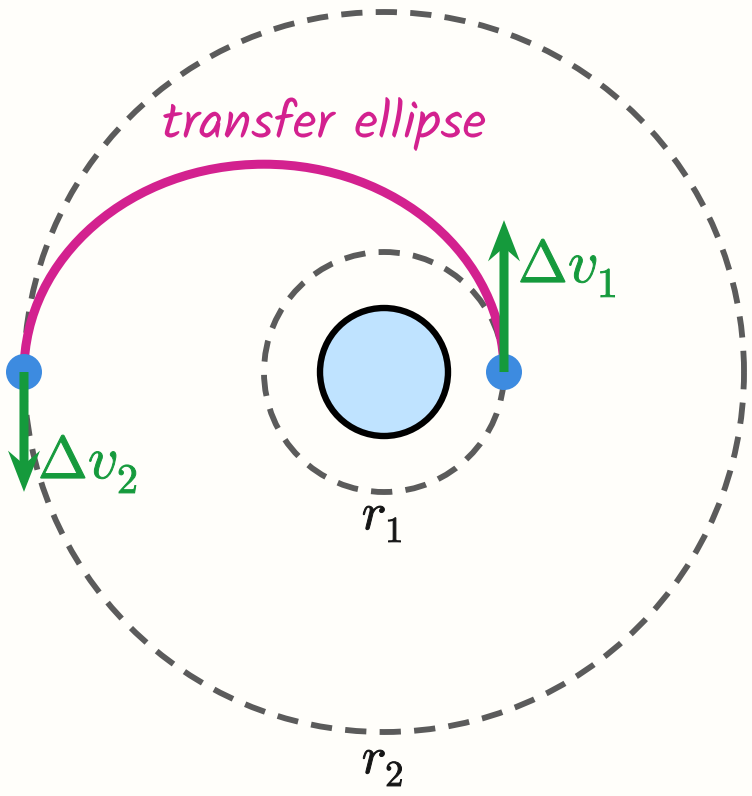
gravity only, Earth fixed ($m \ll M$), no drag, launch velocity $\perp$ radius · drawn to scale: $v^2/v_o^2 = 0.6,\ 1,\ 1.5,\ 2,\ 2.6$

Golden points

★ At the same r: $v_e = \sqrt{2}\,v_o$, so escaping needs only $\approx 41\%$ more speed than circling (but twice the KE).

Transfer Between Orbits

1 Two-burn transfer



- Burn 1 at r_1 (along $\vec{v}$): circle → ellipse with perigee r_1, apogee r_2.
- Burn 2 at r_2 (along $\vec{v}$): ellipse → circle r_2.
- Transfer: $a = \dfrac{r_1 + r_2}{2}$, time $= \frac{1}{2}T_{ellipse} = \pi\sqrt{\dfrac{a^3}{GM}}$.

$$E_1 = -\frac{GMm}{2r_1}, \quad E_t = -\frac{GMm}{r_1 + r_2}, \quad E_2 = -\frac{GMm}{2r_2}$$

circular orbits and the transfer ellipse ($E = -GMm/2a$) · each burn RAISES E · quick burns, $m \ll M$

2 Example: $r_2 = 3r_1$

- Circle speeds: v_1 and $v_2 = v_1/\sqrt{3} \approx 0.577\, v_1$.
- Ellipse (vis-viva): $v_p = v_1\sqrt{2r_2/(r_1 + r_2)} \approx 1.225\, v_1$, $v_a = v_p r_1/r_2 \approx 0.408\, v_1$.
- $\Delta v_1 \approx 0.225\, v_1$, $\Delta v_2 \approx 0.169\, v_1$, both forward.

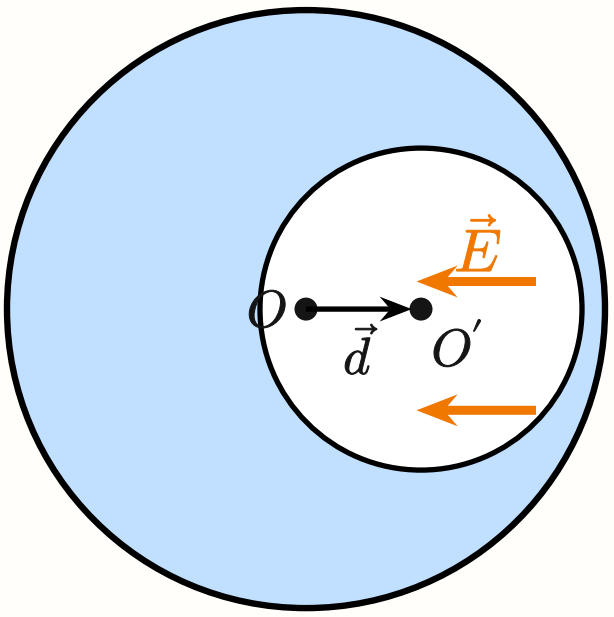
Golden points

★ Going HIGHER: speed up twice, yet end SLOWER in the higher circle.

Cavity Problems

1 Field inside the cavity

- Trick: sphere with a cavity = FULL uniform sphere + a sphere of NEGATIVE mass (same density) filling the cavity (cavity EMPTY).

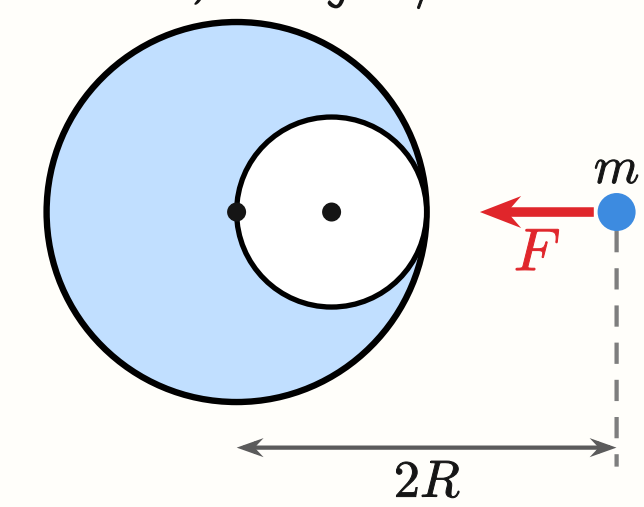


$$\vec{E} = -\frac{4}{3}\pi G\rho\,\vec{d}$$

UNIFORM density ρ, spherical cavity with NO mass inside · $\vec{d}$: from centre O to cavity centre O' · SAME at every point of the cavity, along $O' \to O$

- Proof: inside a uniform sphere $\vec{E} = -\frac{4}{3}\pi G\rho\,\vec{r}$; add the negative sphere's $+\frac{4}{3}\pi G\rho\,(\vec{r} - \vec{d})$.

2 Force on a mass outside



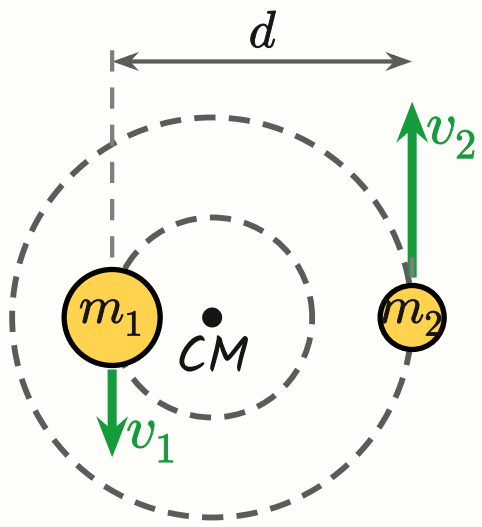
$$F = \frac{GMm}{(2R)^2} - \frac{G(M/8)m}{(1.5R)^2}$$

$$F = \frac{7GMm}{36R^2}$$

M: mass of the FULL sphere (cavity mass $M/8$) · m on the line of centres, cavity side, $2R$ from O

Binary Stars and Two-Body Orbits

1 Both orbit the centre of mass



$$r_1 = \frac{m_2 d}{m_1 + m_2}$$

$$r_2 = \frac{m_1 d}{m_1 + m_2}$$

two bodies, gravity only, circular orbits about the CM · SAME ω and T, always on opposite sides

$$T = 2\pi\sqrt{\frac{d^3}{G(m_1 + m_2)}}$$

from $m_1\omega^2 r_1 = \frac{Gm_1m_2}{d^2}$ · d: separation (not r_1)

2 Reduced mass

$$\mu = \frac{m_1 m_2}{m_1 + m_2}, \quad E = -\frac{Gm_1m_2}{2d}, \quad L = \mu d^2 \omega$$

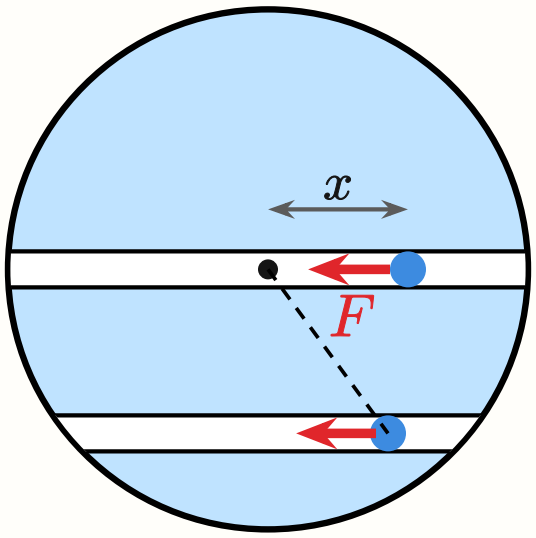
relative motion = one body of mass μ orbiting a FIXED total mass $m_1 + m_2$ · $K_1 : K_2 = m_2 : m_1$

Example

Two equal stars M, separation d: $T = 2\pi\sqrt{d^3/2GM}$, each speed $v = \sqrt{GM/2d}$, radius $d/2$.

Tunnel Through the Earth

1 Diameter tunnel



chord tunnel (lower)

$$F = -\frac{GMm}{R^3}\,x = -\frac{mg}{R}\,x$$

UNIFORM, non-rotating Earth, smooth tunnel, no air · x from the centre · restoring: SHM

$$T = 2\pi\sqrt{\frac{R}{g}} \approx 84.6\,\text{min}$$

$$v_{centre} = \sqrt{gR} \approx 7.92\,\text{km}\,\text{s}^{-1}$$

released from REST at the surface · T equals the period of a satellite skimming the surface

2 Chord tunnel

- Along any chord, the component of F along the tunnel is $-\frac{mg}{R}\,x'$ (x' from the chord's midpoint): SAME $T \approx 84.6\,\text{min}$, for every chord.
- Other end in $T/2 \approx 42.3\,\text{min}$; midpoint in $T/4 \approx 21.2\,\text{min}$.

Golden points

★ Body dropped from REST at height h into the tunnel: not SHM all the way (outside, $F \propto 1/r^2$); use energy with $U = mV(r)$.

Common Traps

1 Exam traps

Avoid these

- ✘ $g = 9.8\,\mathrm{m\,s^{-2}}$ everywhere (only near the surface; g falls with height, with depth for a uniform Earth, and towards the equator for a rotating Earth).
- ✘ $\Delta U = mgh$ for any height (only $h \ll R$; otherwise $GMm\left(\frac{1}{R} - \frac{1}{R+h}\right)$).
- ✘ $g_h \approx g(1-2h/R)$ for any h (only $h \ll R$; exact: $g/(1+h/R)^2$).
- ✘ No gravity in orbit ($\approx 89\%$ of g at $400\,\mathrm{km}$; weightless = free fall).
- ✘ Escape speed depends on the launch angle or the mass (neither, no drag, path not hitting the planet).
- ✘ $E = 0$ inside a shell, so $V = 0$ (uniform thin shell: $V = -GM/R$).
- ✘ Higher orbit needs less energy because it is slower (circular orbits: TOTAL energy $-GMm/2r$ rises with r).
- ✘ $v_p r_p = v_a r_a$ at any two points (only where $\vec{v} \perp \vec{r}$; in general $r\,v\sin\theta$ is conserved).
- ✘ Cavity field is zero (uniform sphere, empty spherical cavity: field is UNIFORM, $\frac{4}{3}\pi G\rho d$).
- ✘ Ellipse period uses r_p or r_a (it uses the SEMI-MAJOR axis $a = (r_p + r_a)/2$, $m \ll M$).

Golden points

★ First ask: inside or outside? circular orbit? is $h \ll R$?

Standard Results

1 One-line results

Quantity	Result	Condition
g at height h	$g/(1+h/R)^2$	spherical Earth, any h
g at depth d	$g(1-d/R)$	uniform Earth
g at latitude λ	$g-\omega^2 R\cos^2\lambda$	spherical, $\omega^2 R \ll g$
Shell (inside)	$E=0,\ V=-GM/R$	uniform thin shell
Solid sphere (inside)	$E=GMr/R^3$	uniform density
Escape speed	$\sqrt{2gR}\approx 11.2\,\mathrm{km\,s^{-1}}$	no drag, no spin
Orbital speed	$\sqrt{GM/r}$	circular orbit
Period	$2\pi\sqrt{r^3/GM}$	circular orbit
Satellite energy	$-GMm/2r$	circular orbit
Geostationary	$h\approx 36000\,\mathrm{km}$	equatorial, $T=24\,\mathrm{h}$

2 Ratios to remember

- $v_e : v_o = \sqrt{2} : 1$ (same r) · $K : U : E = 1 : -2 : -1$ (circular orbit).
- $V_{centre} = 1.5\,V_{surface}$ (uniform solid sphere) · g halves at $d = R/2$ (uniform Earth), quarters at $h = R$.

Summary

$$F=\frac{Gm_1m_2}{r^2} \quad \bullet \quad g=\frac{GM}{R^2} \quad \bullet \quad v_o=\sqrt{\frac{GM}{r}} \quad \bullet \quad T^2\propto a^3$$

Questions: JEE Main Level (I)

$g = 9.8\,\mathrm{m\,s^{-2}}$, $R = 6400\,\mathrm{km}$ (Earth) where numbers are needed

Q1 · JEE Main level · MCQ

At height $h = R/2$ (uniform Earth, no rotation), g is: (A) $4g/9$ (B) $g/4$ (C) $2g/3$ (D) $g/2$

Solution: $g_h = g/(1+1/2)^2 = g/2.25 = 4g/9$ (the $h \ll R$ formula would wrongly give 0).

Answer: (A)

Q2 · JEE Main level · Numerical

Height (in units of R) where g equals its value at depth $R/2$? (uniform Earth, no rotation)

Solution: Depth $R/2$: $g/2$. Height: $g/(1+h/R)^2 = g/2$, $1 + h/R = \sqrt{2}$, $h = (\sqrt{2}-1)R \approx 0.414R$.

Answer: ≈ 0.41

Q3 · JEE Main level · MCQ

The radius of a satellite's circular orbit increases by 1%. Its period: (A) increases by 1.5% (B) increases by 0.5% (C) decreases by 1.5% (D) increases by 3%

Solution: $T \propto r^{3/2}$: $\Delta T/T \approx \frac{3}{2}\Delta r/r = 1.5\%$.

Answer: (A)

Q4 · JEE Main level · Numerical

Planet: $8\times$ Earth's mass, $2\times$ its radius. Its escape speed in $\mathrm{km\,s^{-1}}$? (Earth: 11.2; spherical planets, no drag)

Solution: $v_e \propto \sqrt{M/R} = \sqrt{8/2} = 2$: $2 \times 11.2 = 22.4\,\mathrm{km\,s^{-1}}$.

Answer: 22.4

Questions: JEE Main Level (2)

$g = 9.8\,\mathrm{m\,s^{-2}}$, $R = 6400\,\mathrm{km}$ *(Earth) where numbers are needed*

Q5 · JEE Main level · MCQ

Minimum energy to move a satellite m from a circular orbit of radius $2R$ to one of $4R$: (A) $mgR/8$ (B) $mgR/4$ (C) $mgR/2$ (D) $mgR/16$

Solution: $\Delta E = \frac{1}{2}GMm\left(\frac{1}{2R} - \frac{1}{4R}\right) = GMm/8R = mgR/8$ $(GM = gR^2)$.

Answer: (A)

Q6 · JEE Main level · Numerical

Perihelion $1.0 \times 10^{11}\,\mathrm{m}$ at $30\,\mathrm{km\,s^{-1}}$; aphelion $1.5 \times 10^{11}\,\mathrm{m}$. Speed at aphelion $(\mathrm{km\,s^{-1}})$?

Solution: L *conserved,* $\vec{v} \perp \vec{r}$ *at both:* $v_a = 30 \times 1.0/1.5 = 20\,\mathrm{km\,s^{-1}}$.

Answer: 20

Q7 · JEE Main level · MCQ

Length of a day if bodies at the equator felt no weight? (spherical Earth) (A) $\approx 1.4\,\mathrm{h}$ (B) $\approx 24\,\mathrm{h}$ (C) $\approx 12\,\mathrm{h}$ (D) $\approx 5.1\,\mathrm{h}$

Solution: $\omega^2 R = g$: $T = 2\pi\sqrt{R/g} \approx 5077\,\mathrm{s} \approx 1.41\,\mathrm{h}$.

Answer: (A)

Q8 · JEE Main level · Numerical

Minimum energy $(10^{10}\,\mathrm{J})$ to put $1000\,\mathrm{kg}$ from rest on the surface into a circular orbit at $h = R$? (non-rotating Earth, no drag)

Solution: $E_{orbit} - U_{surf} = -\frac{1}{4}mgR + mgR = \frac{3}{4}mgR \approx 4.70 \times 10^{10}\,\mathrm{J}$.

Answer: ≈ 4.7

Questions: JEE Advanced Level (I)

A1 · JEE Advanced level · One or more correct

A uniform solid sphere has mass M and radius R ($V=0$ at infinity). Correct: (A) $V_{centre} = 1.5\,V_{surface}$ (B) $E(R/2) = 2E(2R)$ (C) work to move m slowly from the centre to the surface is $\dfrac{GMm}{2R}$ (D) the field is largest at the centre

Solution: $V_c = -\dfrac{3GM}{2R}$, $V_s = -\dfrac{GM}{R}$: (A) true. $E(R/2) = \dfrac{GM}{2R^2}$, $E(2R) = \dfrac{GM}{4R^2}$: (B) true. $W = m(V_s - V_c) = \dfrac{GMm}{2R}$: (C) true. $E = 0$ at the centre: (D) false.

Answer: (A), (B), (C)

A2 · JEE Advanced level · Numerical

From a uniform sphere (mass M before cutting, radius R) a sphere of radius $R/2$ touching the surface is removed (cavity empty). A mass m sits on the line of centres, on the cavity side, at $2R$ from the centre. $F = k\dfrac{GMm}{R^2}$; find k.

Solution: Cavity mass $M/8$ at $1.5R$ from m:

$$F = \frac{GMm}{4R^2} - \frac{GMm}{8\times 2.25R^2} = \frac{GMm}{R^2}\left(\frac{1}{4} - \frac{1}{18}\right) = \frac{7}{36}\frac{GMm}{R^2}.$$

Answer: $k = 7/36 \approx 0.194$

Tip

★ Cavity: compute for the full sphere, then SUBTRACT the cavity treated as a point mass at its centre (outside points) or with the $\vec{r}$ formula (inside points).

Questions: JEE Advanced Level (2)

gravity only, $m \ll M$ where a planet is the centre

A3 · JEE Advanced level · Paragraph

Two stars of masses M and $2M$, separation d, move in circles about their centre of mass. Find (i) the period, (ii) the ratio of kinetic energies $K_M : K_{2M}$, (iii) the total energy.

Solution: (i) $T = 2\pi\sqrt{\dfrac{d^3}{G(3M)}}$. (ii) Same ω, $r_M = \dfrac{2d}{3}$, $r_{2M} = \dfrac{d}{3}$; $K \propto mr^2$, $K_M : K_{2M} = M \cdot \frac{4}{9} : 2M \cdot \frac{1}{9} = 2:1$. (iii) $E = -\dfrac{G(M)(2M)}{2d} = -\dfrac{GM^2}{d}$.

Answer: (i) $2\pi\sqrt{d^3/3GM}$ (ii) $2:1$ (iii) $-GM^2/d$

A4 · JEE Advanced level · One or more correct

A satellite at distance r_0 from the Earth's centre is given a speed $v = \sqrt{\dfrac{4GM}{3r_0}}$ perpendicular to the radius (T_0: period of the circle of radius r_0). Correct: (A) the launch point is the perigee (B) greatest distance $2r_0$ (C) speed at the farthest point $v/2$ (D) period $2\sqrt{2}\,T_0$

Solution: $v > \sqrt{GM/r_0}$: perigee, (A). $E = \dfrac{m}{2}\cdot\dfrac{4GM}{3r_0} - \dfrac{GMm}{r_0} = -\dfrac{GMm}{3r_0} = -\dfrac{GMm}{2a}$; $a = 1.5r_0$, $r_{max} = 2a - r_0 = 2r_0$, (B). L: $v_{min} = v\dfrac{r_0}{2r_0} = \dfrac{v}{2}$, (C). $T = (1.5)^{3/2}T_0 \approx 1.84\,T_0$, (D) false.

Answer: (A), (B), (C)

Questions: JEE Advanced Level (3)

uniform, non-rotating Earth, no air drag, $g = 9.8\,\mathrm{m\,s^{-2}}$, $R = 6400\,\mathrm{km}$

AS · JEE Advanced level · Numerical

A body is released from rest at height R above the surface, directly over a smooth tunnel along a diameter. Its speed (in $\mathrm{km\,s^{-1}}$) at the Earth's centre?

Solution: $V(2R) = -\dfrac{GM}{2R}$, $V(0) = -\dfrac{3GM}{2R}$; $\frac{1}{2}v^2 = \dfrac{GM}{R} = gR$, $v = \sqrt{2gR} = \sqrt{2 \times 9.8 \times 6.4 \times 10^6} = 11.2\,\mathrm{km\,s^{-1}}$ (equal to v_e!).

Answer: 11.2

A6 · JEE Advanced level · Matching

Satellite in a CIRCULAR orbit of radius r. Match List I with how it depends on r (List II). (P) orbital speed (Q) period (R) angular momentum (S) magnitude of total energy. (1) $r^{1/2}$ (2) r^{-1} (3) $r^{-1/2}$ (4) $r^{3/2}$

Solution: $v = \sqrt{GM/r}$; $T = 2\pi\sqrt{r^3/GM}$; $L = m\sqrt{GMr}$; $|E| = GMm/2r$.

Answer: P-3, Q-4, R-1, S-2

How to use these

- ★ Try each question before reading its solution; redo the ones you missed after a day.
- ★ Check every answer a second way (energy and angular momentum, or a limiting case such as $h \ll R$).